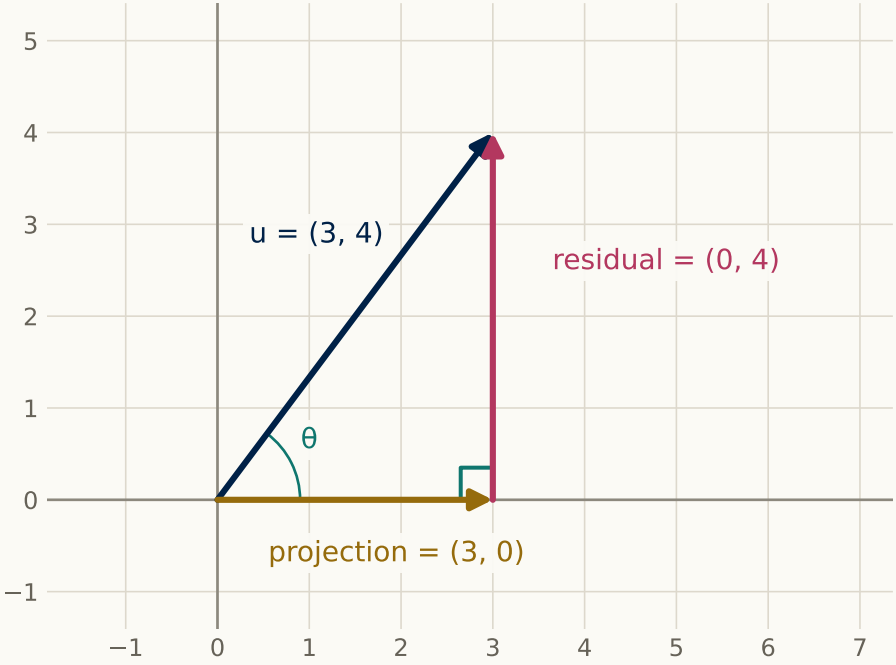Projection leaves a perpendicular residual